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Objective	Motivated Software Engineering student with a passion for Cyber-Security and AI , seeking fully funded research or software engineering / cyber-security internships to contribute to high-impact projects and accelerate my professional growth.
Education	FAST NUCES, Pakistan — BS in Software Engineering Aug 2022 – Present Pakistan Scouts Cadet College Batrasi — Intermediate in Pre-Engineering 2020 – 2022
Experience	Research Assistant – Cybersecurity Awareness, FAST NUCES Jan 25 – Present - Analyzed student cybersecurity behavior and contributed to awareness initiatives and research publication. - Designed and conducted surveys with 500+ responses; co-authored publication currently under review. Intern – Data Science, 10Pearls April 26 – Present - Applied data analysis and machine learning techniques to real-world datasets. Intern – DevOps & Automation, SPS July 25 – November 25 - Improved project tracking efficiency by leveraging Jira in cross-functional teams. - Automated deployment workflows, reducing time-to-market by 20%. - Worked with cloud infrastructure and containerization to build scalable solutions. Intern – AI Enhanced Web Development, Nexium July 25 – Aug 25 - Developed web applications using Next.js and React, focusing on performance and user experience. Intern – Cybersecurity, Velocity Business Solutions June 25 – July 25 - Performed vulnerability analysis and worked with industry-standard security tools. - Applied offensive and defensive security concepts in practical projects.
Final Year Project	DeceptiCloud: AI-Driven Cyber Deception System Using Adaptive Honeypots in Cloud Environment Aug 2025 – Present Technologies: Python, PyTorch (WGAN-GP), TensorFlow/Keras, Scikit-learn, Flask, SQLite, Docker, Kubernetes, Wazuh SIEM, Ollama (Gemma-3), GeoIP2, JA3 Fingerprinting, Nginx - Architected a cloud-native cyber deception platform deploying 14 adaptive honeypots (7 production replicas + 7 mirror clones: banking, e-commerce, healthcare, REST API, corporate, blog, admin panel) behind an AI-driven reverse proxy that silently redirects adversarial traffic to isolated deception environments with zero attacker awareness. - Engineered a 7-model ML detection pipeline: Keras DNN for web attacks (SQLi/NoSQLi/XSS), Random Forest DDoS detector (>99% balanced accuracy), dedicated RF classifiers for XSS (TF-IDF), brute-force, port-scan, and credential-stuffing, plus Isolation Forest for zero-day anomalies — served via a unified Flask ML API with hot-reload after retraining. - Implemented an ensemble decision layer fusing rule-based scoring with ML confidence; integrated a Layer-7 HTTP DDoS detector using per-IP sliding-window features (req/s, UA entropy, error rate) to detect application-layer floods, and a WGAN-GP (PyTorch) synthetic data generator for honeypot seeding and ML data augmentation with mode-collapse early stopping. - Developed an LLM Adaptive Response Engine (Ollama/Gemma-3 1B) generating contextually convincing deception HTML (fake SQL dumps, XSS reflections, shell output) via a fast-seed + background HQ cache strategy; engineered a Canary Token system (12+ endpoints: .env, backup.sql, wp-admin, /.git) with active defenses — a ZIP bomb (2.9 KB → 10 TB, Fifield 2019) and an EICAR scare trap to incapacitate automated scanners. - Implemented a blockchain-backed immutable attack ledger (SHA-256 chained blocks, Proof-of-Work) for forensic-grade tamper-proof logging; designed a multi-factor behavioral fingerprinting engine (canvas/WebGL hashes, keystroke biometrics, mouse dynamics, JA3 pseudo-TLS, MaxMind GeoIP2) with DBSCAN-inspired clustering to correlate attackers across VPN/Tor pivots. - Built a Behavioral Monitor cross-referencing real-site traffic against known-attacker fingerprints (behavioral 40% + JA3 30% + GeoIP 20% + IP 10%) for session termination at ≥50% similarity; developed an Attacker Profiling Engine computing weighted threat scores and grouping actors via K-Means for SOC-ready attribution. - Integrated a Model Drift Detector monitoring 6-hour confidence windows — triggering auto-retraining on >10% confidence drop, >25% uncertainty ratio, or ≥200 new samples; integrated Wazuh SIEM with custom XML decoders and 15+ correlation rules for sub-second correlated incident response feeding back into attacker threat scores. - Deployed 17+ containerized microservices via Docker and Kubernetes with an SSH honeypot (port 2222), persistent IP blocklist, and a 10-table normalized SQLite DB; built a real-time analytics dashboard (Flask, 14 API modules) surfacing live attack feeds, cluster cards, canary alerts, LLM metrics, blockchain integrity, and model drift reports for full SOC situational awareness.
Technical Skills	Languages: Rust, Python, C++, C, Java, MySQL, MongoDB Web: HTML5, CSS3, JavaScript, PHP Frameworks: TensorFlow, PyTorch, Scikit-learn, Flask, FastAPI, Bootstrap, jQuery, EcmaScript, Vanilla.js, MITRE ATT&CK, Metasploit Tools: Git, Linux, Docker, Kubernetes, Wazuh SIEM, Jira, Eclipse, PyCharm, IntelliJ IDEA, Figma, Ettercap, Wireshark, Nmap, AWS Soft Skills: Problem-solving, Teamwork, Leadership, Adaptability, Learning Agility, Time Management
Projects	Auto-Mechanics – Web-Based Vehicle Repair Platform: PHP, MySQL, and JavaScript implementation. Hotel Management System: Object-oriented principles demonstration in C++. Message Broadcaster & Group Chat: Real-time messaging system using socket programming. Connect-Sphere – Community Engagement App: Built with Flask and Tkinter. Explorer App – UI/UX Design: Clean, user-centered interface enhancing usability. RK2/RK4 Simulation: Predator-prey modeling in Python. AI-Driven Matrimonial Platform: Smart matchmaking with lightweight ML model and chatbot support. Blog Summarizer & Translator: Full-stack app for content scraping, summarization, and translation. Multi-Agent Stock Trading System: AI simulation using Naive Bayes, K-Means, and multi-agent framework. Secure Authentication System: CSRF protection, rate limiting, password hashing, session management. WebSec-Assessment Tool: Automated scanning for XSS, SQL injection, ports, and SSL/TLS vulnerabilities.
Certifications & Achievements	Certified Cyber Security, Google Certified DevOps, Udemy Self Alignment to Startup, FAST-NUCES Evolving Tech & Project Management, FAST Structured LLM Output, deeplearning.ai Rust Programming Course: From Beginner to Expert 2.0, Udemy
Extracurriculars	Director Operations – MLSA Team Get it Done Member – GDGoC ACM FAST Chapter – Member Canva - Graphic Designing and Editing Volunteer – Zindagi Prize Vice Captain Hockey Team – Cadet College
Languages	English: B2 Urdu: C1 Hindko: Native